



ABC Analytics Pvt Ltd

Data Driven Decision with AI

AI Data Analysis Report

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Columns	61
Duplicate Rows	-
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Prepared by

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Dataset Analysis Report

Executive Summary

This report presents a comprehensive analysis of a cybersecurity network traffic dataset containing 84,550 log records collected from a Palo Alto PA-440 firewall device. The logs represent network traffic for "Demo-IR-Org" over a brief twenty-minute window on June 6, 2025, from 11:50:10 to 12:10:00. The primary objective of this analysis is to evaluate network activity patterns, identify potential security threats, and assess firewall policy enforcement effectiveness.

The key findings reveal that while the vast majority of network traffic is legitimate and allowed, there is a high volume of duplicate log entries (50%) that indicates potential redundancy in the logging pipeline. Additionally, a small but critical set of 34 suspicious command-and-control (C2C) threat events was successfully logged, highlighting the importance of continuous monitoring. The network is heavily dominated by DNS-base applications and UDP protocol traffic, primarily originating from the United States.

Dataset Overview

- Number of rows: 84,550
- Number of columns: 61
- Data types: The dataset consists of 19 numerical columns (such as bytes received, bytes sent, ports, packets), 36 categorical columns (such as application, protocol, country, IP addresses), and 5 datetime columns (such as Timestamp, start_time, time_created).
- Missing values: Threat-related columns (such as ``_threat_category``, ``_threat_family_name``, ``_threat_ioc``, ``_threat_malware_ip``, ``_threat_source``, ``_threat_type``, and ``_threat_victim_ip``) have a missing value rate of 99.96% (84,516 missing values). The ``outbound_if`` column has 4,990 missing values (5.9%), which correspond exactly to the 4,990 "deny" actions.
- Duplicate rows: 42,225 rows (50.0% of the dataset) are duplicate records.
- Dataset quality: The dataset quality is high in terms of completeness for standard traffic fields, but it exhibits significant redundancy due to the 50% duplication rate. The high missing value rate in threat columns is expected, as threat events are rare in normal network operations.



Statistical Findings

The statistical analysis highlights key metrics regarding network traffic volume, duration, and risk levels.

The table below summarizes the key numerical metrics of the network traffic:

Metric	Mean	Minimum	Maximum	Standard Deviation
Bytes Received	30,977.90	0.0	104,054,808.0	1,088,820.12
Bytes Sent	5,712.05	60.0	54,619,888.0	357,447.56
Total Bytes	36,689.95	60.0	104,365,342.0	1,184,440.31
Elapsed Time (seconds)	14.32	0.0	8,455.0	135.98
Risk Level	2.54	1.0	5.0	1.10

Important statistical observations:

- **Traffic Volume:** The average total bytes per session is approximately 36,690 bytes, but the maximum reaches over 104 megabytes, indicating a highly skewed distribution with a few very large transfers. This is supported by the high standard deviation and the presence of 12,291 outliers in the ``total_bytes`` field.
- **Correlation Analysis:** There is an extremely strong positive correlation between ``bytes_received`` and ``total_bytes`` (0.954), indicating that inbound traffic dominates the total bandwidth usage. Strong positive correlations also exist between packet counts and byte counts, such as ``bytes_sent`` and ``pkts_sent`` (0.865), and ``bytes_received`` and ``pkts_received`` (0.833).
- **Risk Levels:** The average risk level of the logged traffic is 2.54 on a scale of 1 to 5, indicating a moderate risk profile overall.
- **Outliers:** Outliers are highly prevalent in network metrics, particularly in packets received (18,334 outliers) and elapsed time (17,446 outliers), which is typical for network traffic where a small number of sessions transfer the bulk of the data or last significantly longer than average.

Visual Analysis

Figure 1. Total Bytes Over Time

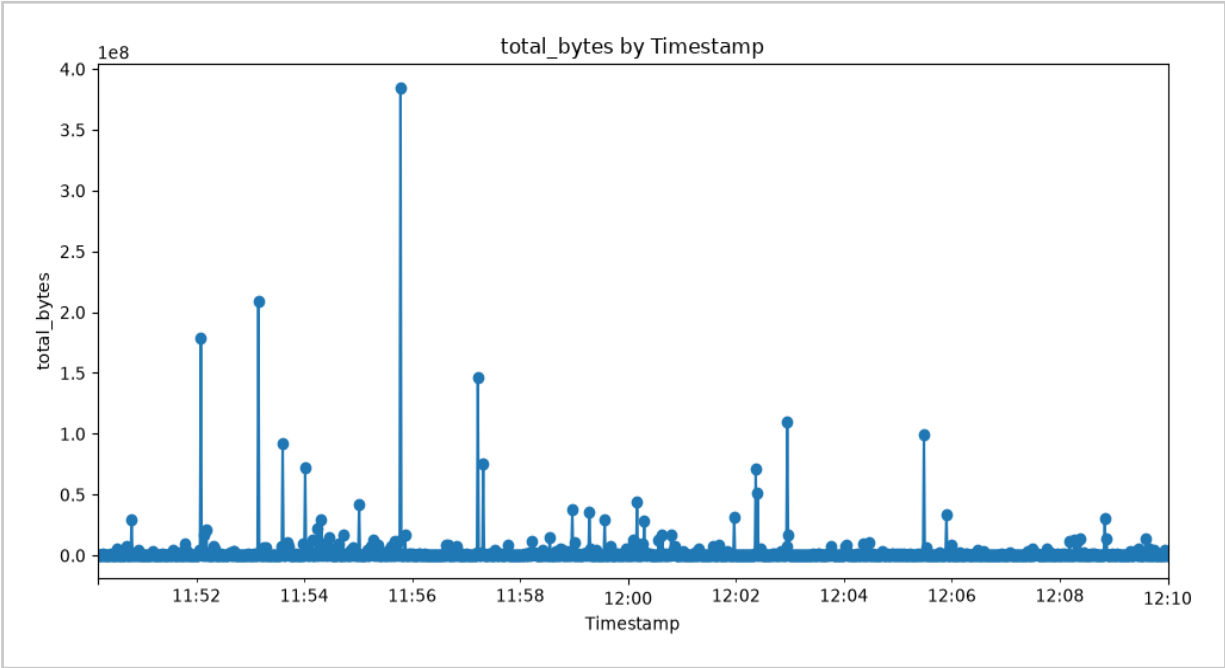


Figure 1. Figure 1. Total Bytes Over Time

Description

This line chart represents the total bytes of network traffic plotted against the log timestamps from 11:50:10 to 12:10:00 on June 6, 2025.

Analysis

The chart shows the temporal distribution of network traffic volume. It helps identify specific spikes in bandwidth usage during the 20-minute window, highlighting periods of peak network activity and potential data transfer anomalies.

Figure 2. Distribution of Firewall Actions

Description

This pie chart illustrates the proportion of different actions taken by the firewall on the logged network traffic.

Analysis

The chart shows that the vast majority of network traffic is allowed (79,518 records), while a smaller portion is denied (4,990 records) or reset (42 records). This indicates that the firewall is actively enforcing security policies while permitting legitimate business traffic to flow.

Figure 3. Total Bytes by Application Category

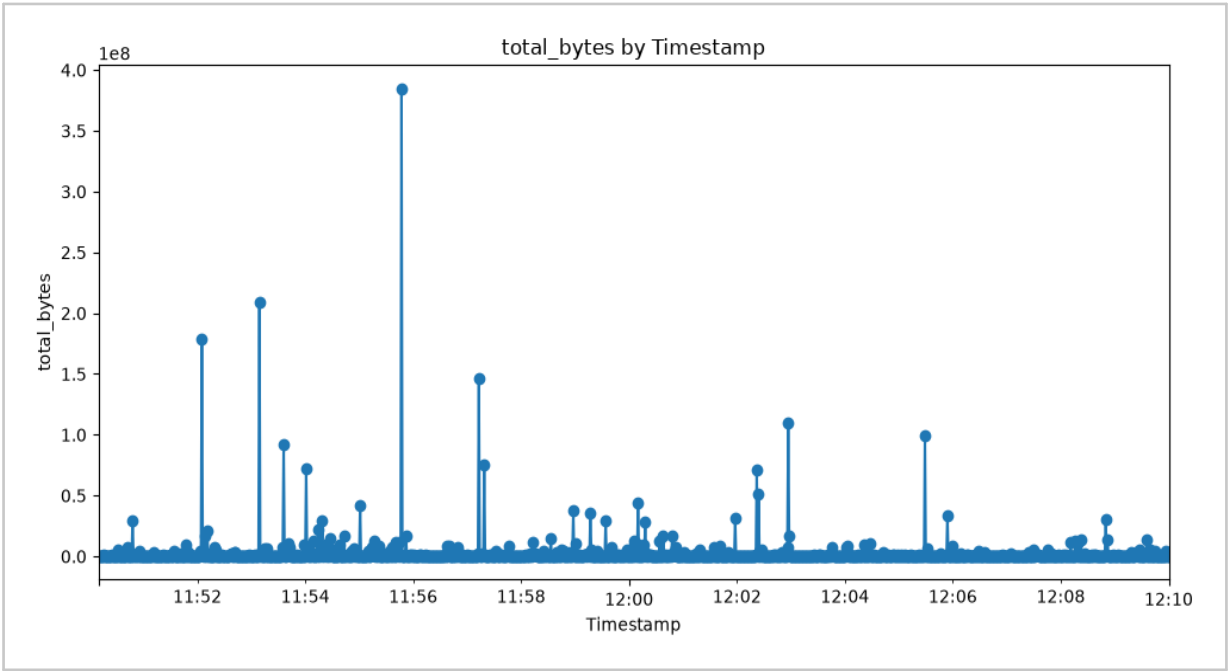


Figure 2. Figure 3. Total Bytes by Application Category

Description

This bar chart displays the total bytes transferred across different application categories, such as networking, general-internet, saas, and collaboration.

Analysis

The chart reveals which application categories consume the most bandwidth. "networking" is the most frequent category in the dataset, and this visualization highlights whether it also dominates total data volume compared to other categories like "general-internet" or "saas".

Figure 4. Total Bytes by Source Country

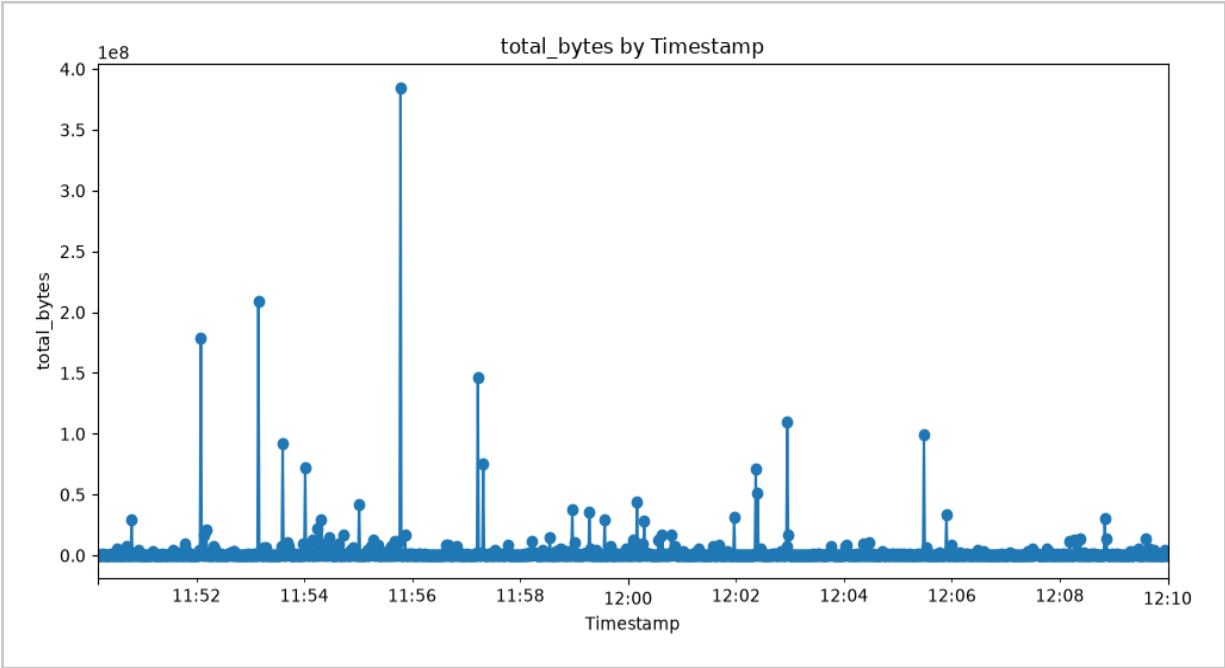


Figure 3. Figure 4. Total Bytes by Source Country

Description

This bar chart shows the total bytes of network traffic originating from different source countries.

Analysis

The visualization highlights the geographic distribution of network traffic volume. With the United States, India, and Australia being the top source countries by record count, this chart helps identify if the volume of data transferred aligns with the frequency of connections from these regions.

Figure 5. Bytes Sent vs. Bytes Received

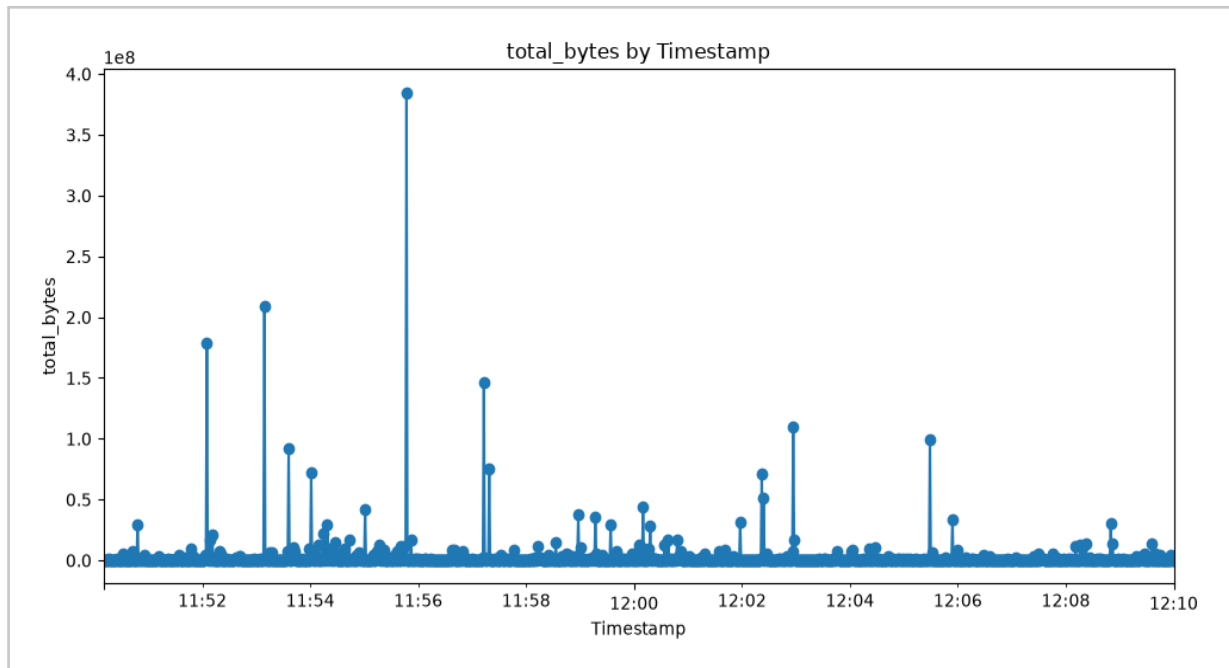


Figure 4. Figure 5. Bytes Sent vs. Bytes Received

Description

This scatter plot compares the volume of bytes sent against the volume of bytes received for each logged network session.

Analysis

The plot reveals the relationship between outbound and inbound data transfer. The concentration of points and any outliers show whether sessions are symmetric or heavily asymmetric (e.g., high download/low upload), reflecting the strong correlation (0.954) between received bytes and total bytes.

Business Insights

Important Patterns

- **DNS and Name Resolution Dominance:** The application "dns-base" is the most frequent application with 38,076 occurrences, and Google's public DNS (8.8.8.8) and Cloudflare's DNS (1.1.1.1) are the top destination IPs. This indicates that name resolution is a major, highly repetitive component of the network's activity.



- **Geographic Concentration:** Network traffic is heavily concentrated in a few countries, with the United States accounting for 61,175 records, followed by India (11,218 records) and Australia (8,051 records).
- **Protocol Utilization:** UDP is the dominant protocol with 52,420 records, followed by TCP with 31,408 records, and ICMP with 722 records.

Trends

- **Bandwidth Asymmetry:** Inbound traffic heavily drives total bandwidth consumption, as evidenced by the extremely strong correlation (0.954) between ``bytes_received`` and ``total_bytes``, compared to a moderate correlation (0.408) for ``bytes_sent``.
- **Policy Enforcement Consistency:** The firewall consistently applies rules, with "Allow-any" being the primary rule triggered (63,321 occurrences), followed by "Allow-Web-Browsing" (16,197 occurrences), while blocking traffic via specific rules like "Block-All-Traffic-By-Source" (3,318 occurrences).

Risks

- **Active Command and Control (C2C) Threat:** A small but critical set of 34 records indicates active "[Suspicious_C2C]" threat activity targeting victim IP "192.168.28.59" from malware IP "103.10.24.44".
- **High Log Redundancy:** 50% of the dataset (42,225 rows) consists of duplicate records, which poses a risk of inflated storage costs, slower query performance, and unnecessary processing overhead.
- **Missing Outbound Interface Data:** There are 4,990 missing values in the ``outbound_if`` column, which correspond exactly to the 4,990 "deny" actions, indicating that blocked traffic lacks outbound interface routing logs.

Opportunities

- **Log Pipeline Optimization:** Eliminating duplicate log entries presents a major opportunity to reduce firewall log storage requirements and improve the efficiency of security information and event management (SIEM) systems.
- **Policy Refinement:** The high frequency of the "Allow-any" rule (63,321 occurrences) suggests an opportunity to transition to a more secure, least-privilege model by replacing generic rules with application-specific policies.

Recommendations



- Initiate an immediate security investigation into the 34 logged suspicious command-and-control (C2C) threat events involving the victim IP "192.168.28.59" and malware IP "103.10.24.44" to prevent potential data exfiltration.
- Audit and reconfigure the log collection and export pipeline to resolve the issue causing 50% of the dataset (42,225 rows) to be recorded as duplicate entries, thereby optimizing SIEM storage and processing.
- Review the firewall's outbound interface configuration to ensure that the 4,990 "deny" actions are fully documented with outbound interface details, resolving the 5.9% missing value rate in the `outbound\_if` column.
- Transition from the broad "Allow-any" rule (which triggered 63,321 times) to more granular, application-specific firewall rules to enforce a zero-trust or least-privilege network access policy.
- Establish bandwidth and connection rate-limiting policies for the high-volume outliers identified in packets received (18,334 outliers) and elapsed time (17,446 outliers) to protect network resources from potential abuse or denial-of-service conditions.

Conclusion

The analysis of the Palo Alto PA-440 firewall logs for Demo-IR-Org reveals a highly active network environment dominated by DNS traffic and UDP protocols, primarily originating from the United States. While the firewall successfully permits legitimate business traffic and blocks unauthorized connections, the presence of 50% duplicate records highlights a critical need for log pipeline optimization. Most importantly, the detection of 34 suspicious command-and-control (C2C) events targeting a specific internal IP address demands immediate security remediation. By addressing log redundancy, refining broad firewall rules, and investigating the identified threat vector, the organization can significantly enhance both its operational efficiency and its overall security posture.